Experiment 7: Configure a local Kubernetes cluster using Minikube and deploy an Nginx Pod using a YAML file.

Kubectl: is command line tool for Kubernetes where we can directly interact with Kubernetes

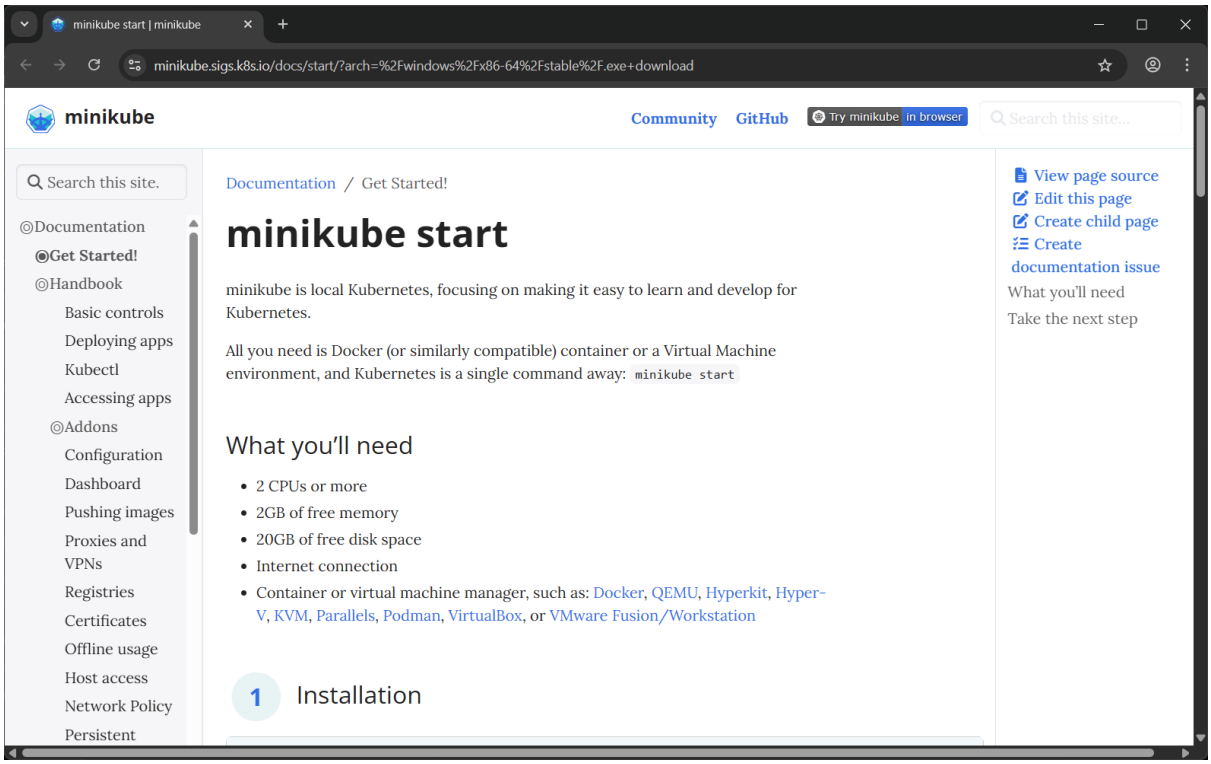
Minikube: is a local Kubernetes cluster

Before that start the docker desktop engine

Step 1: Minikube installation

Go to bowser type minikube installation

https://minikube.sigs.k8s.io/docs/start/?arch=%2Fwindows%2Fx86-64%2Fstable%2F.exe+download

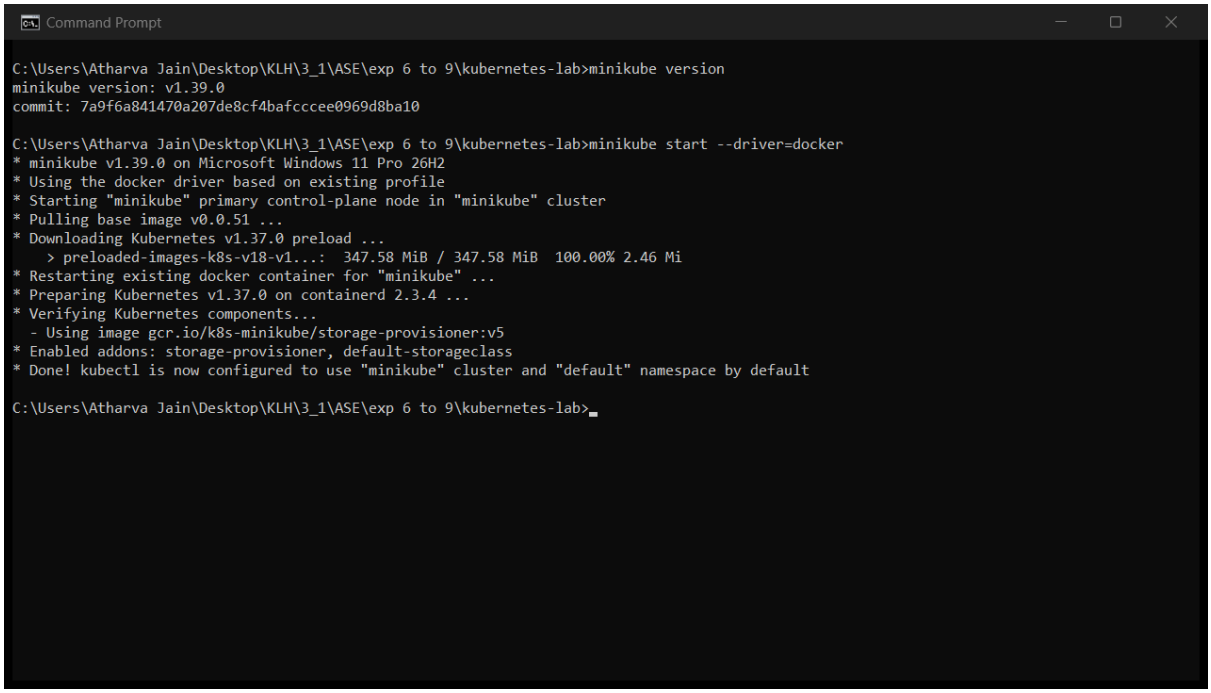


Click on latest release

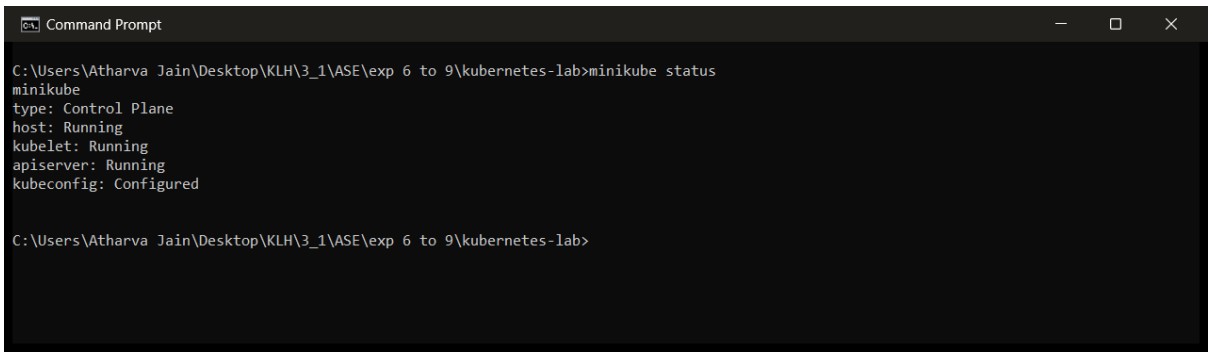
Cmd->run as administrator

>minikube

>minikube start --driver=docker



>minikube status



If you get the error for the above commands
minikube stop

minikube delete

minikube start --driver=docker

OR

minikube delete

docker version

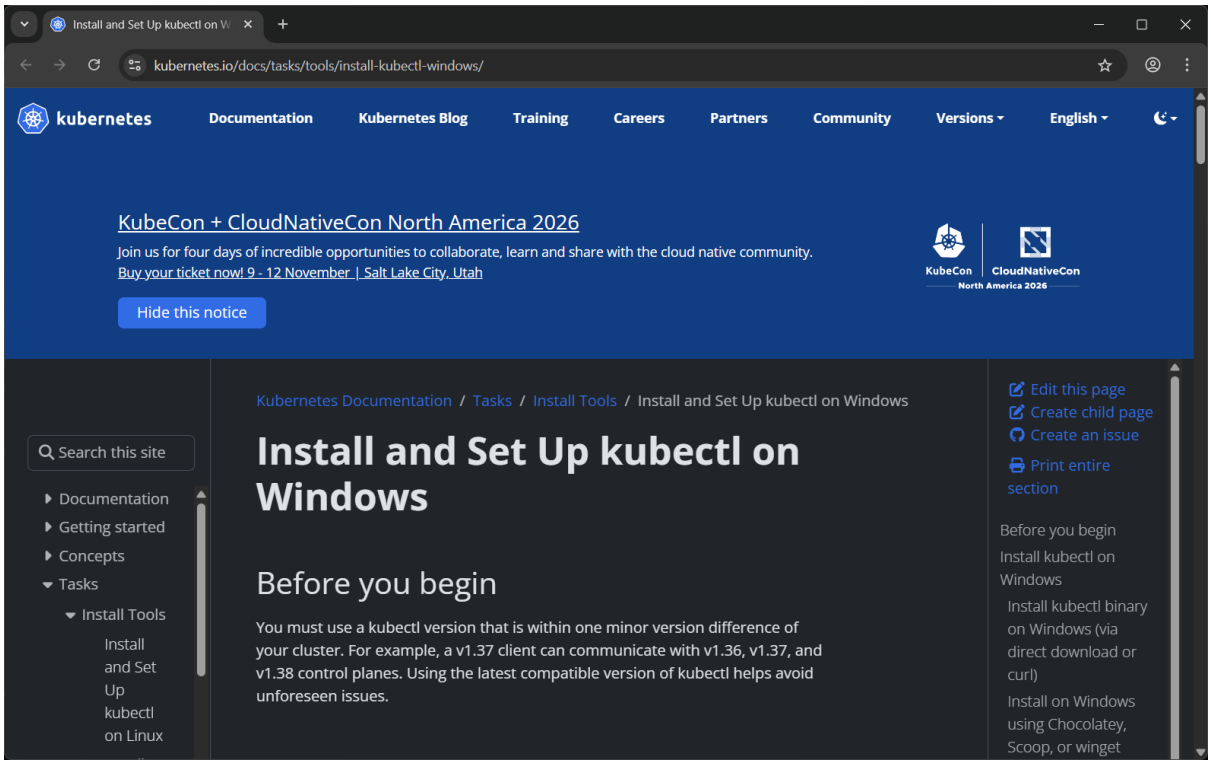
docker info

minikube start --driver=docker

Step 2: Installation of Kubectl

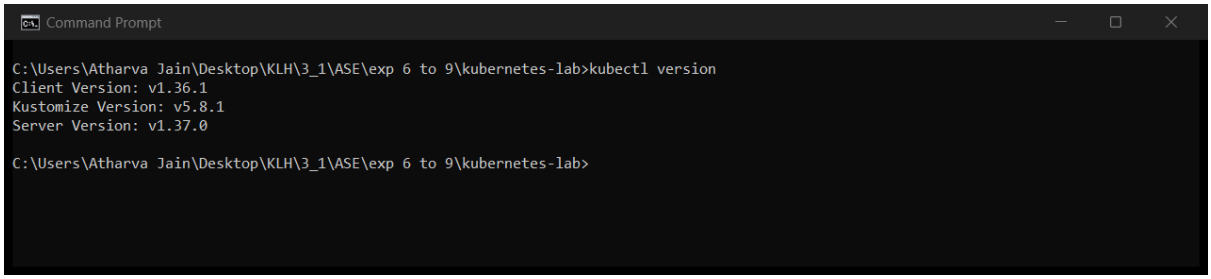
Type kubectl installation-->click on install tools-->choose platform

https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/



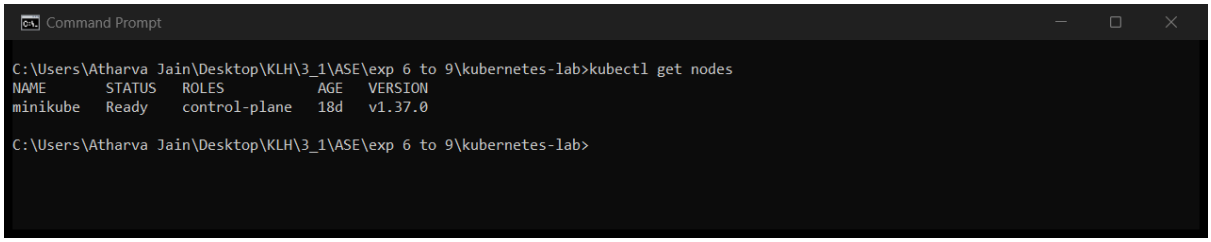
Copy the Curl and paste in the cmd

>kubectl version



Check whether kubectl is connected with minikube by the command

>kubectl get nodes



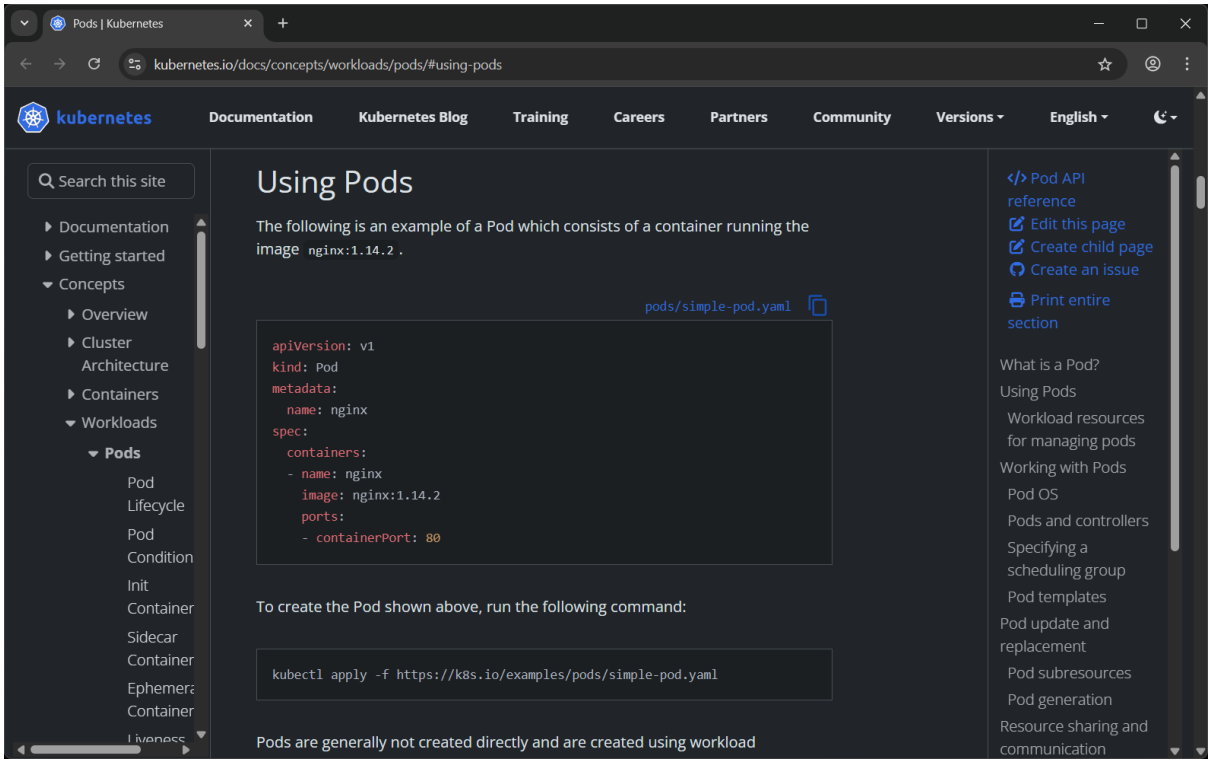
Step 3: Installation of POD

Goto web bowser and type Kubernetes pod

https://kubernetes.io/docs/concepts/workloads/pods/

Pod is basically a Yaml file

Copy the yaml code



Cmd>vi pod.yml

Paste the code (if vi is not recognized)

Cmd> notepad pod.yaml

Click yes

Copy the code and Save the file and close Notepad.

Untitle • | index.html | Dockerfile | pod.y ×

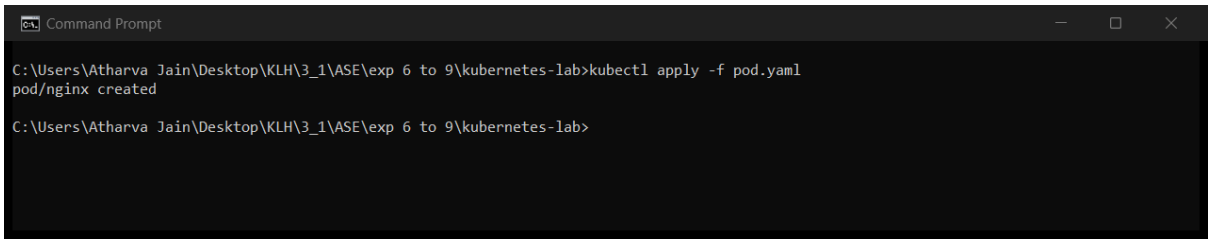
File Edit View

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

Ln 1, Col 1 | 144 characters | Plain text | 100% | Unix (LF) | UTF-8

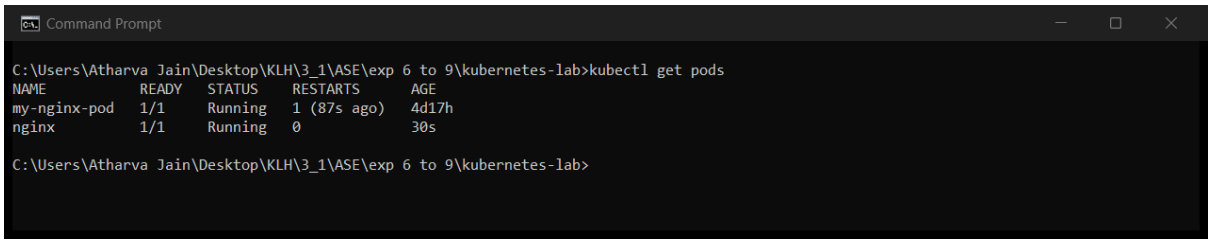
Deploy it:

Cmd> kubectl apply -f pod.yaml



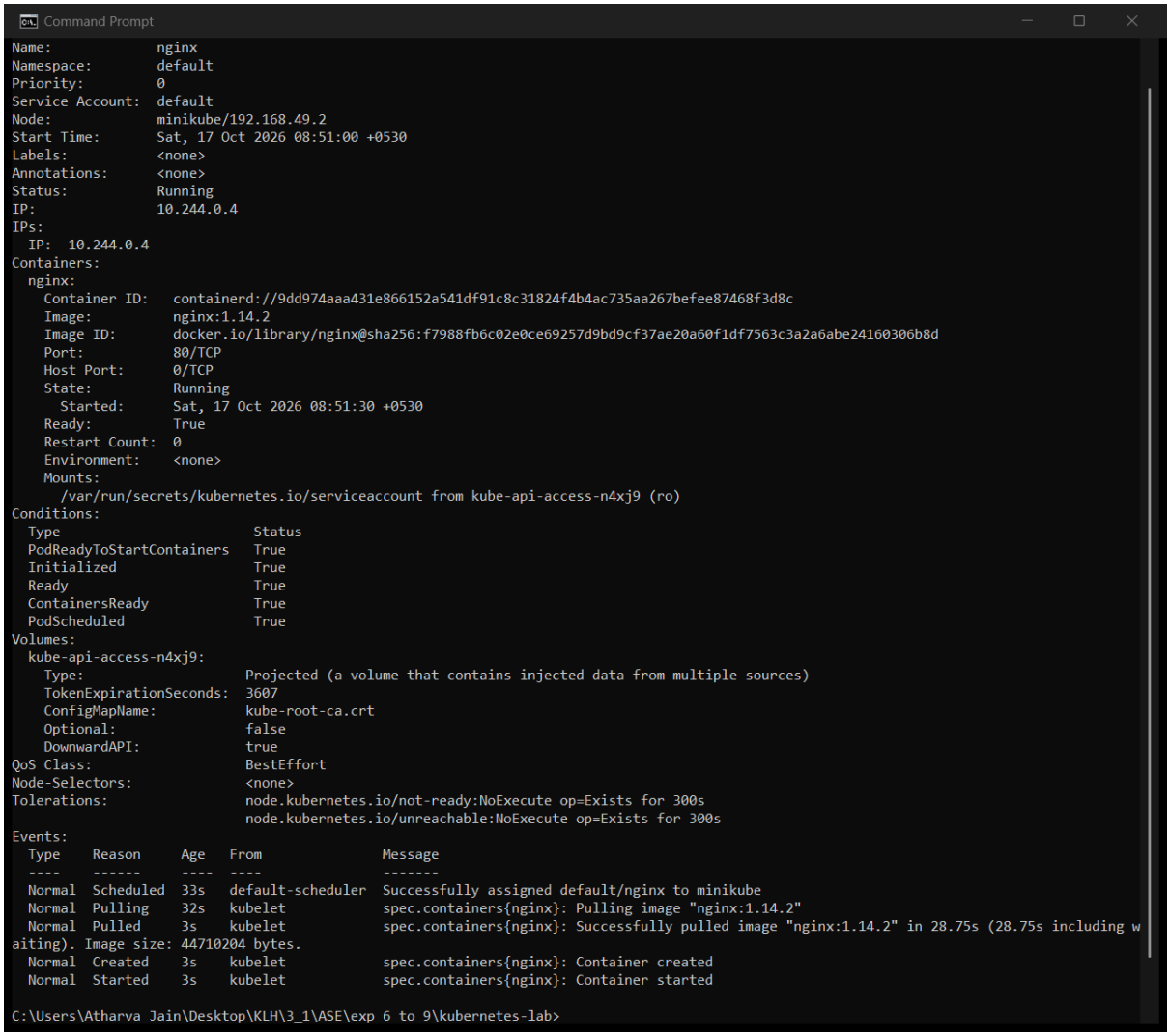
Verify the Pod

> kubectl get pods



Get Detailed Information

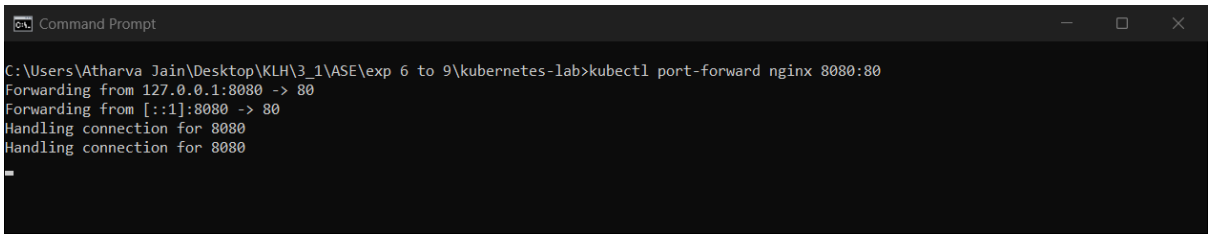
> kubectl describe pod nginx



Access the Pod

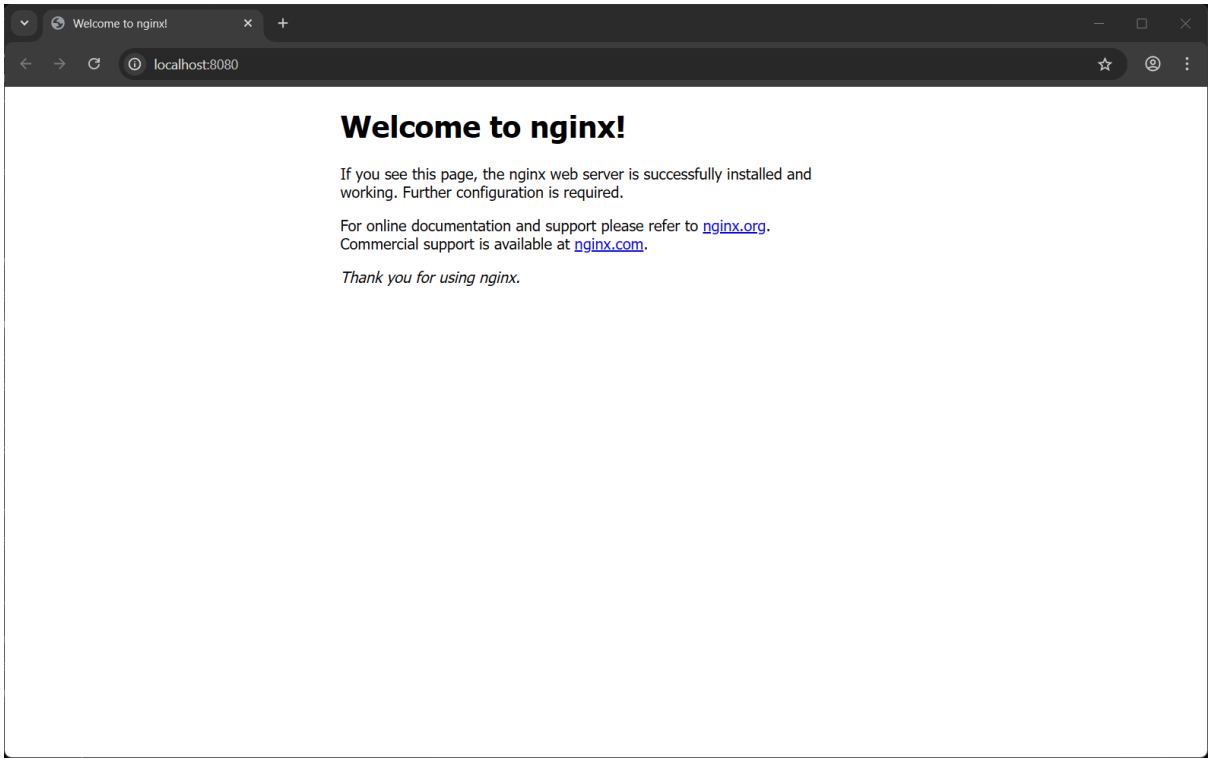
kubectl port-forward nginx 8080:80

Output



Open a browser and visit:

http://localhost:8080


Welcome to nginx!
localhost:8080
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.